HCLTech | FUTURE READY AI ENGINEER HACKATHON

AI-Powered Smart Library Management Assistant

Problem Statement and Participant Use Case

MISSION

Build an accurate, explainable and secure AI assistant that transforms the university library into an intelligent knowledge hub for discovery, research, circulation and planning.

Use case	Smart Library Management
Target users	Students, faculty, researchers and librarians
Solution focus	RAG, semantic search, recommendations, analytics and workflow automation
Delivery model	Hackathon prototype with measurable outcomes

Classification: Internal

1. Problem Statement

University libraries manage books, journals, research papers, e-books, dissertations and digital resources across multiple repositories. Students often struggle to find relevant materials, confirm availability, track due dates and discover content aligned to their courses or research interests.

Librarians spend substantial effort resolving repetitive queries, maintaining catalog quality, monitoring circulation, handling overdue items and identifying future resource demand.

Your Mission

> **Build an AI-powered Smart Library Management Assistant that helps users discover, access, manage and use library resources efficiently through intelligent search, personalized recommendations, workflow automation and analytics.**

Business Challenges

Challenge	Operational impact
Difficulty locating relevant resources	Slower learning and research progress
Repetitive manual assistance	High librarian workload and inconsistent response times
Underused physical and digital collections	Low return on subscriptions and acquisitions
Overdue items and renewal queries	Administrative effort and user dissatisfaction
Limited demand visibility	Reactive acquisition and stock planning
Generic discovery experience	Lower engagement and missed relevant content

2. Functional Requirements

Capability	Required behavior	Example
Intelligent search	Search books, journals, papers, dissertations and e-resources using natural language and metadata.	Find beginner-friendly books on machine learning.
Personalized recommendations	Recommend resources based on course context, research interests and permitted usage history.	Suggest resources for my Data Science semester.
Availability and reservation	Show availability, branch or shelf location, waitlist and reservation status.	Is the required AI textbook available?
Account management	Display borrowed items, due dates, renewability and applicable penalties.	Which borrowed books are due this week?
Research assistance	Summarize authorized content, surface related papers and organize references.	Compare approaches used in these three papers.
Librarian analytics	Identify resource demand, circulation patterns, overdue trends and catalog gaps.	Which titles have the longest waitlists?

Core principle

Use generative AI to understand intent and explain results. Use authoritative library systems and deterministic tools for availability, reservations, loans, fines and renewals. The model must never invent a title, availability status or account transaction.

3. Example User Questions

Question type	Example question	Expected system action
Resource search	Find books on Python programming.	Semantic catalog search with ranked, cited results.
Availability	Is Artificial Intelligence: A Modern Approach available?	Call the catalog/holdings tool and return current status.
Recommendation	Suggest beginner resources for data science.	Generate an explainable recommendation list.
Personal account	What items have I borrowed?	Authenticate the user and retrieve only their circulation data.
Research support	Find papers related to retrieval-augmented generation.	Search authorized repositories and show source metadata.
Unanswerable	Find a title not present in any authorized source.	Return not_found rather than fabricating a result.

AI and Data Components

Layer	Components
Data sources	Library catalog, circulation system, digital repository, journal subscriptions, university research archive and authorized learning systems.
AI capabilities	Semantic search, RAG, summarization, recommendation, conversational assistance and anomaly detection.
Data engineering	Metadata harmonization, document ingestion, deduplication, indexing, vector storage, access control and data quality validation.
Analytics	Demand forecasting, circulation trends, resource utilization, waitlist patterns and overdue analysis.

4. Suggested Solution Architecture

1. Experience	2. Orchestration	3. Intelligence	4. Systems
Web or mobile chat Library portal Librarian dashboard	API gateway Identity and authorization Workflow/router	RAG and semantic search Recommendation engine Analytics models	Catalog and circulation Digital repository Vector DB and audit store

Key Design Controls

- Grounding: Answer resource and policy questions only from authorized sources.
- Deterministic transactions: Use tools for reservations, renewals, availability and charges.
- Identity and privacy: Restrict account data to the authenticated user and role.
- Explainability: Show why each resource was recommended and where it came from.
- Freshness: Reflect catalog and circulation updates without hard-coded responses.
- Auditability: Capture retrieved sources, tools invoked, outcomes, latency and errors.

Example End-to-End Flow

User intent	Find an available beginner book on Python and reserve it.
Search	Semantic catalog search identifies relevant beginner-level titles.
Validation	Holdings tool checks branch, copy and current availability.
Action	Reservation tool creates the hold after authorization.
Response	Assistant confirms the selected title, location and reservation result with a trace ID.

5. Expected Outcomes and Sample Interaction

Outcome area	Expected benefit	How to measure
Resource discovery	Faster, more relevant search experience.	Search success rate, time to useful result, click-through.
Service productivity	Fewer repetitive queries handled manually.	Deflection rate and librarian handling time.
Collection utilization	Greater use of relevant physical and digital assets.	Circulation, downloads and unique users.
Research support	Quicker identification and synthesis of sources.	Task completion and citation accuracy.
Inventory planning	Improved demand visibility for acquisitions.	Waitlist duration, unmet demand and forecast error.

Sample Interaction

STUDENT

Recommend beginner resources for learning generative AI that are currently available.

ASSISTANT

Recommended resources
1. Introductory book - available in Central Library
2. E-book - accessible through the university portal
3. Learning path - available in the digital repository

Why these: beginner level, aligned to the stated topic, and confirmed through authorized holdings.

Next action: Open, reserve or save to reading list.

6. Hackathon Success Criteria

Criterion	What judges should see
Grounded discovery	Relevant search results with catalog or repository evidence; no invented resources.
Tool-based transactions	Availability, reservation, renewal and account results come from deterministic tools.
Personalization	Recommendations are relevant, explainable and privacy-aware.
Live data handling	New or updated resources become searchable without code changes.
Responsible AI	Authorization, abstention, untrusted-content handling and audit controls are demonstrated.
Evaluation	A measured test set covers search relevance, citation accuracy, tool correctness, privacy and latency.
Engineering quality	Clean architecture, documented APIs, repeatable setup and traceable contributions.

Minimum Demonstration Scenario

- One natural-language resource search with relevant ranked results.
- One real-time availability check using a catalog tool.
- One authorized reservation or renewal workflow.
- One personalized recommendation with an explanation.
- One not_found response for unsupported content.
- One privacy refusal for another user's borrowing data.
- One librarian analytics view showing demand or utilization patterns.

TARGET END STATE

A secure Intelligent Knowledge Hub that combines high-quality discovery, research assistance, personalized recommendations and data-driven library operations.